

Math 122: Exam 1 Practice 2 Solutions

Problem 1

Find the exact arc length of the curve $y = (2x + 1)^{3/2}$ over the interval $[0, 1]$.

Solution:

The formula for arc length L from $x = a$ to $x = b$ is given by:

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

First, find the derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{3}{2}(2x + 1)^{1/2} \cdot 2 = 3(2x + 1)^{1/2}$$

Square the derivative:

$$\left(\frac{dy}{dx}\right)^2 = (3(2x + 1)^{1/2})^2 = 9(2x + 1) = 18x + 9$$

Substitute into the arc length integrand:

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + 18x + 9 = 18x + 10$$

Set up and evaluate the integral:

$$L = \int_0^1 \sqrt{18x + 10} dx$$

Let $u = 18x + 10$, so $du = 18 dx \implies dx = \frac{du}{18}$.

- When $x = 0$, $u = 10$.
- When $x = 1$, $u = 28$.

$$L = \frac{1}{18} \int_{10}^{28} u^{1/2} du = \frac{1}{18} \left[\frac{2}{3} u^{3/2} \right]_{10}^{28} = \frac{1}{27} (28^{3/2} - 10^{3/2})$$

Simplifying $28^{3/2} = 28\sqrt{28} = 56\sqrt{7}$ and $10^{3/2} = 10\sqrt{10}$:

$$L = \frac{56\sqrt{7} - 10\sqrt{10}}{27} \approx 4.32$$

Problem 2

Find the limit:

$$\lim_{x \rightarrow 0} \frac{\tan(2x)}{\ln(1+x)}$$

Solution:

Direct substitution gives:

$$\frac{\tan(0)}{\ln(1)} = \frac{0}{0}$$

This is an indeterminate form of type $\left[\frac{0}{0}\right]$, so we apply L'Hôpital's Rule:

$$\lim_{x \rightarrow 0} \frac{\tan(2x)}{\ln(1+x)} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx}[\tan(2x)]}{\frac{d}{dx}[\ln(1+x)]} = \lim_{x \rightarrow 0} \frac{2 \sec^2(2x)}{\frac{1}{1+x}}$$

Evaluating the limit by direct substitution:

$$= \frac{2 \sec^2(0)}{\frac{1}{1+0}} = \frac{2(1)}{1} = 2$$

Problem 3

Find the limit:

$$\lim_{x \rightarrow +\infty} x^2 e^{-2x}$$

Solution:

Rewrite the limit as a quotient:

$$\lim_{x \rightarrow +\infty} \frac{x^2}{e^{2x}}$$

Direct substitution yields $[\frac{\infty}{\infty}]$. Applying L'Hôpital's Rule:

$$\lim_{x \rightarrow +\infty} \frac{\frac{d}{dx}[x^2]}{\frac{d}{dx}[e^{2x}]} = \lim_{x \rightarrow +\infty} \frac{2x}{2e^{2x}} = \lim_{x \rightarrow +\infty} \frac{x}{e^{2x}}$$

This is still in the indeterminate form $[\frac{\infty}{\infty}]$. Applying L'Hôpital's Rule a second time:

$$\lim_{x \rightarrow +\infty} \frac{\frac{d}{dx}[x]}{\frac{d}{dx}[e^{2x}]} = \lim_{x \rightarrow +\infty} \frac{1}{2e^{2x}} = 0$$

Problem 4

Find the limit:

$$\lim_{x \rightarrow 0^+} (1 + 2x)^{1/x}$$

Solution:

This is an indeterminate form of type $[1^\infty]$. Let $y = (1 + 2x)^{1/x}$.

Take the natural logarithm of both sides:

$$\ln y = \ln((1 + 2x)^{1/x}) = \frac{\ln(1 + 2x)}{x}$$

Find $\lim_{x \rightarrow 0^+} \ln y$:

$$\lim_{x \rightarrow 0^+} \frac{\ln(1 + 2x)}{x}$$

Direct substitution gives $[\frac{0}{0}]$. Applying L'Hôpital's Rule:

$$\lim_{x \rightarrow 0^+} \frac{\frac{2}{1+2x}}{1} = \frac{2}{1+0} = 2$$

Since $\lim_{x \rightarrow 0^+} \ln y = 2$, exponentiate to find the original limit:

$$\lim_{x \rightarrow 0^+} y = e^2$$

Problem 5

Find $\frac{dy}{dx}$ for $y = \cos^{-1}\left(\frac{1}{x^2}\right)$.

Solution:

Using the derivative rule $\frac{d}{dx}[\cos^{-1}(u)] = -\frac{1}{\sqrt{1-u^2}} \cdot \frac{du}{dx}$.

Let $u = \frac{1}{x^2} = x^{-2}$, so $\frac{du}{dx} = -2x^{-3} = -\frac{2}{x^3}$.

$$\begin{aligned}\frac{dy}{dx} &= -\frac{1}{\sqrt{1-\left(\frac{1}{x^2}\right)^2}} \cdot \left(-\frac{2}{x^3}\right) \\ \frac{dy}{dx} &= \frac{2}{x^3 \sqrt{1-\frac{1}{x^4}}} = \frac{2}{x^3 \sqrt{\frac{x^4-1}{x^4}}} \\ \frac{dy}{dx} &= \frac{2}{x^3 \cdot \frac{\sqrt{x^4-1}}{x^2}} = \frac{2}{x \sqrt{x^4-1}}\end{aligned}$$

Problem 6

Find $\frac{dy}{dx}$ for $y = \sin^{-1} x \cdot \cos^{-1} x$.

Solution:

Apply the Product Rule: $\frac{d}{dx}[u \cdot v] = u'v + uv'$.

Recall:

$$\frac{d}{dx}[\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}[\cos^{-1} x] = -\frac{1}{\sqrt{1-x^2}}$$

Applying the rule:

$$\begin{aligned}\frac{dy}{dx} &= \left(\frac{1}{\sqrt{1-x^2}}\right) \cdot \cos^{-1} x + \sin^{-1} x \cdot \left(-\frac{1}{\sqrt{1-x^2}}\right) \\ \frac{dy}{dx} &= \frac{\cos^{-1} x - \sin^{-1} x}{\sqrt{1-x^2}}\end{aligned}$$

Problem 7

Evaluate the integral:

$$\int \frac{dx}{1+9x^2}$$

Solution:

Rewrite the integrand in the standard inverse tangent form $\int \frac{du}{1+u^2} = \tan^{-1} u + C$:

Let $u = 3x$, then $du = 3 dx \implies dx = \frac{du}{3}$.

$$\int \frac{dx}{1+(3x)^2} = \frac{1}{3} \int \frac{du}{1+u^2} = \frac{1}{3} \tan^{-1} u + C = \frac{1}{3} \tan^{-1}(3x) + C$$

Problem 8

Evaluate the integral:

$$\int \frac{x dx}{\sqrt{1-x^4}}$$

Solution:

Rewrite x^4 as $(x^2)^2$:

$$\int \frac{x dx}{\sqrt{1-(x^2)^2}}$$

Let $u = x^2$, then $du = 2x dx \implies x dx = \frac{du}{2}$.

$$\int \frac{\frac{1}{2} du}{\sqrt{1-u^2}} = \frac{1}{2} \sin^{-1} u + C = \frac{1}{2} \sin^{-1}(x^2) + C$$

Problem 9

Evaluate the definite integral:

$$\int_0^1 \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}} dx$$

Solution:

Let $u = \sin^{-1} x$, then $du = \frac{1}{\sqrt{1-x^2}} dx$.

Change the limits of integration:

- When $x = 0$, $u = \sin^{-1}(0) = 0$.
- When $x = 1$, $u = \sin^{-1}(1) = \frac{\pi}{2}$.

Substitute into the integral:

$$\int_0^{\pi/2} u^3 du = \left[\frac{u^4}{4} \right]_0^{\pi/2} = \frac{(\pi/2)^4}{4} - 0 = \frac{\pi^4/16}{4} = \frac{\pi^4}{64} \approx 1.52$$

Problem 10

Evaluate the integral:

$$\int \frac{e^x}{1 + e^{2x}} dx$$

Solution:

Rewrite e^{2x} as $(e^x)^2$:

$$\int \frac{e^x}{1 + (e^x)^2} dx$$

Let $u = e^x$, then $du = e^x dx$.

Substitute into the integral:

$$\int \frac{du}{1 + u^2} = \tan^{-1} u + C = \tan^{-1}(e^x) + C$$